

VCF Private Cloud Deployment – Network Architecture Overview

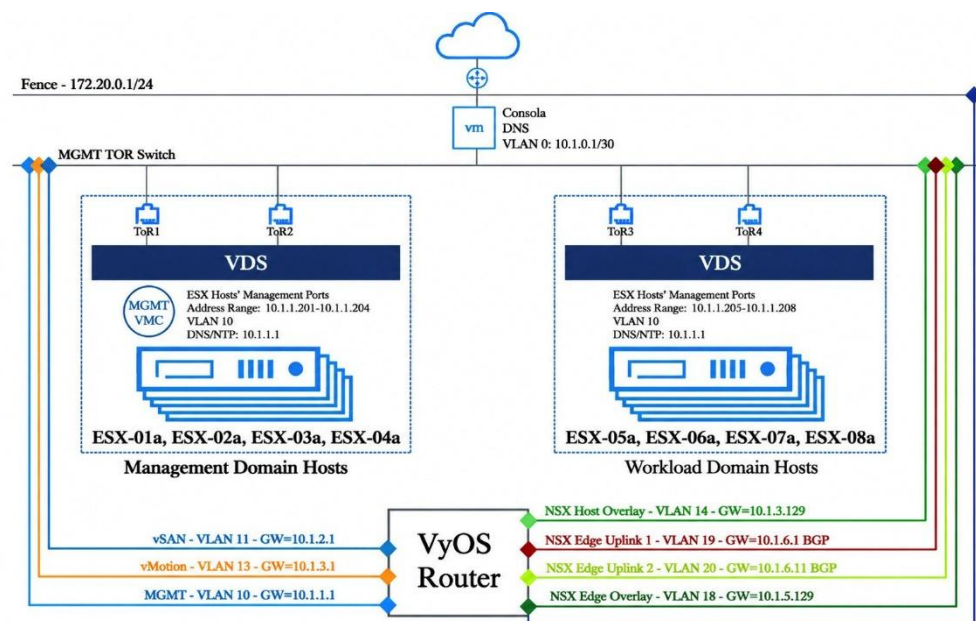
The VCF Private Cloud environment is built on a **redundant network** architecture consisting of management and workload domains connected through a pair of Top-of-Rack (ToR) switches and a VyOS router.

The **Management Domain** includes four ESXi hosts (ESX-01a to ESX-04a), while the **Workload Domain** consists of four additional ESXi hosts (ESX-05a to ESX-08a). Both domains use dedicated Virtual Distributed Switches (VDS).

The network is segmented into dedicated VLANs for **Management, vSAN, vMotion, and NSX overlay/edge traffic**. A VyOS router provides the required gateways and routing between these networks, including BGP connectivity for the NSX Edge uplinks.

A dedicated **DNS/Console VM** provides DNS and management access for the environment. The management network uses VLAN 10 (10.1.1.0/24), while vSAN and vMotion use VLANs 11 and 13, respectively. NSX traffic is isolated across dedicated overlay and Edge VLANs.

This architecture provides the **connectivity, network segmentation, redundancy, and routing foundation required for the VCF Private Cloud deployment**.



VCF Private Cloud Logical Architecture

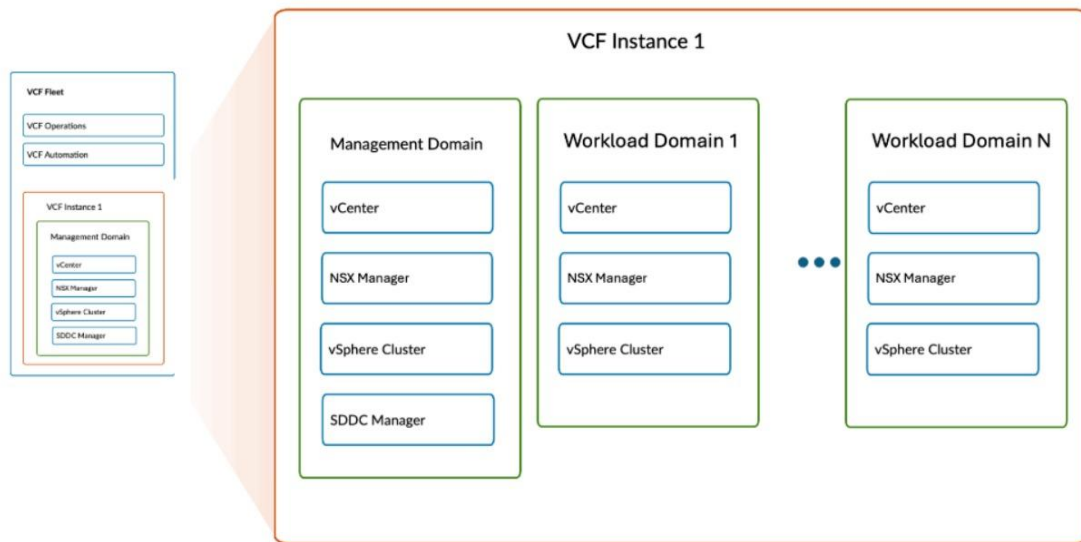
The **VCF Private Cloud** is organized into a **Management Domain** and one or more **Workload Domains** within a VCF Instance.

The **Management Domain** hosts the core VCF infrastructure, including **vCenter**, **NSX Manager**, the **vSphere Cluster**, and **SDDC Manager**. Workload Domains provide dedicated environments for running tenant and application workloads, with each domain containing its own **vCenter**, **NSX Manager**, and **vSphere Cluster**.

At the platform level, **VCF Fleet** provides centralized management capabilities through **VCF Operations** and **VCF Automation**, enabling consistent operations and management across the VCF environment.

Full Image

×



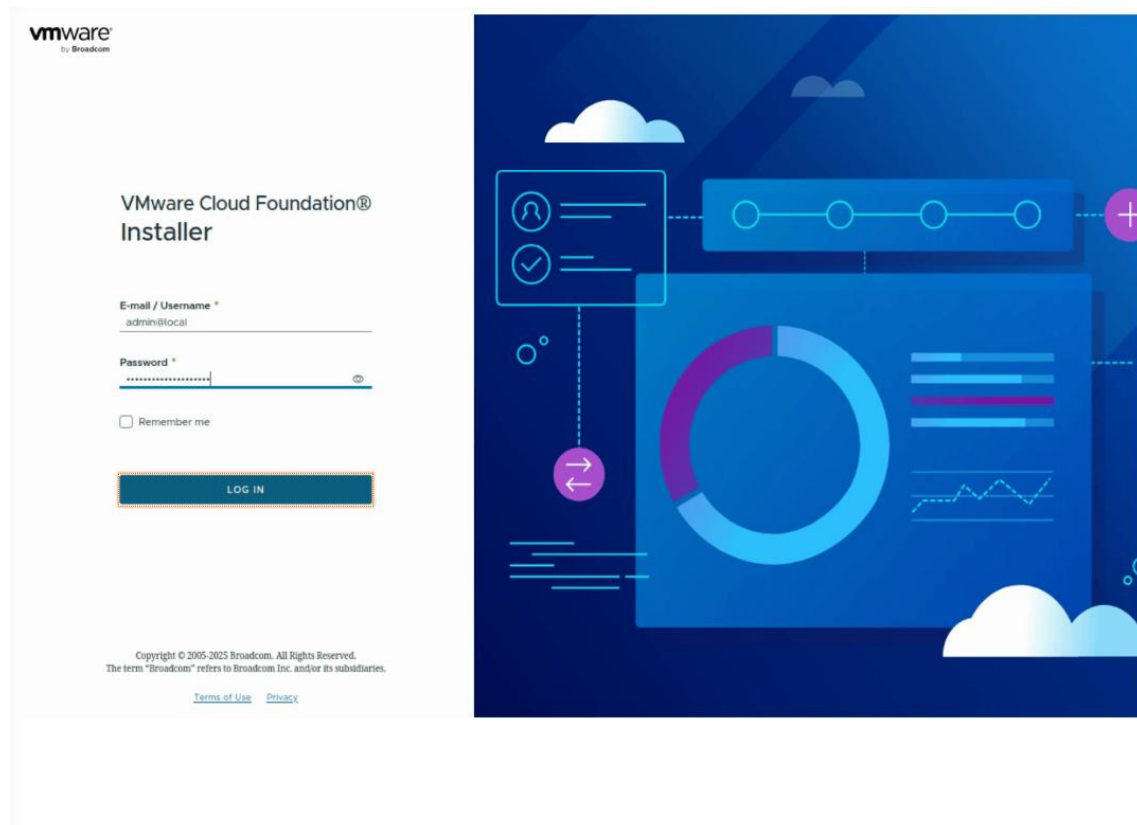
VMware Cloud Foundation Installer

Initial Login

After deploying the VMware Cloud Foundation (VCF) Installer virtual appliance, navigate to the IP address of the VCF Installer appliance from a browser. The default username of 'admin@local' for initial login should auto populate.

To login:

1. Click to enter the password.
2. Click on the '**LOG IN**' button.

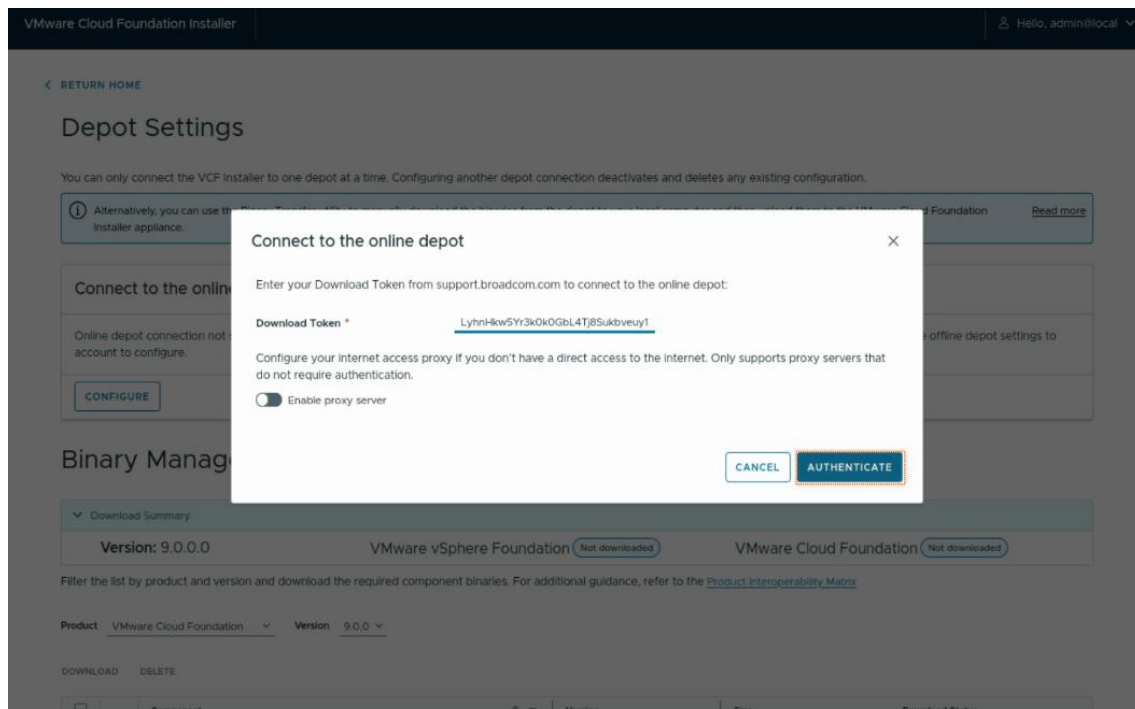


Configure Online Depot Connection

The first step upon login is to set up the VCF depot connection and download the required software binaries. VCF Installer appliance ships without software binaries. You will need your Broadcom download token that you can obtain from support.broadcom.com.

1. Click '**DEPOT SETTINGS AND BINARY MANAGEMENT**'.
2. Click '**CONFIGURE**' to enter your Broadcom entitlement token.
3. To enter your download token, click the Download Token text field. For the purpose of this demonstration, the token will automatically be entered.
4. Click '**AUTHENTICATE**' to verify and save your token.

The online depot connection is now active.



Download Software Binaries

We will now download the required software binaries to install VCF.

1. Click on the vertical scroll bar to the right to scroll down.
2. We will be downloading all the required software for the pre-selected VCF Version 9.0.0. Click on the checkbox on the top left of the table to select all.
3. Click '**DOWNLOAD**' to start downloading from the configured online depot completed in previous steps.

The required software binaries will be downloaded, validated and show a status of 'Success' when complete.

VMware Cloud Foundation Installer

Hello, admin@local

EDIT DEPOT CONNECTIONDISCONNECT

CONFIGURE

Binary Management

Download Summary

Version: 9.0.0.0VMware vSphere FoundationNot downloadedVMware Cloud FoundationNot downloaded

Filter the list by product and version and download the required component binaries. For additional guidance, refer to the [Product Interoperability Matrix](#)

Product VMware Cloud FoundationVersion 9.0.0

DOWNLOADDELETE

☒	Component	Version	Size	Download Status
☒	SDDC Manager	9.0.0.0	2.00 GB	Not downloaded
☒	VMware Cloud Foundation Automation	9.0.0.0	21.17 GB	Not downloaded
☒	VMware Cloud Foundation Operations	9.0.0.0	2.63 GB	Not downloaded
☒	VMware Cloud Foundation Operations Collector	9.0.0.0	2.68 GB	Not downloaded
☒	VMware Cloud Foundation Operations fleet management	9.0.0.0	1.49 GB	Not downloaded
☒	VMware NSX	9.0.0.0	9.99 GB	Not downloaded
☒	VMware vCenter	9.0.0.0	12.04 GB	Not downloaded

VMware Cloud Foundation Installer

Hello, admin@local

EDIT DEPOT CONNECTIONDISCONNECT

CONFIGURE

Binary Management

Download Summary

Version: 9.0.0.0VMware vSphere FoundationDownloadedVMware Cloud FoundationDownloaded

Filter the list by product and version and download the required component binaries. For additional guidance, refer to the [Product Interoperability Matrix](#)

Product VMware Cloud FoundationVersion 9.0.0

DOWNLOADDELETE

☐	Component	Version	Size	Download Status
☐	SDDC Manager	9.0.0.0	2.00 GB	Success
☐	VMware Cloud Foundation Automation	9.0.0.0	21.17 GB	Success
☐	VMware Cloud Foundation Operations	9.0.0.0	2.63 GB	Success
☐	VMware Cloud Foundation Operations Collector	9.0.0.0	2.68 GB	Success
☐	VMware Cloud Foundation Operations fleet management	9.0.0.0	1.49 GB	Success
☐	VMware NSX	9.0.0.0	9.99 GB	Success
☐	VMware vCenter	9.0.0.0	12.04 GB	Success

1. Scroll up the window by clicking on the top right scroll bar.
2. Click '**RETURN HOME**' to return to the main screen for the next step.

VMware Cloud Foundation Installer
Hello, admin@local

[RETURN HOME](#)

Depot Settings

You can only connect the VCF Installer to one depot at a time. Configuring another depot connection deactivates and deletes any existing configuration.

Alternatively, you can use the Binary Transfer utility to manually download the binaries from the depot to your local computer and then upload them to the VMware Cloud Foundation Installer appliance. [Read more](#)

Connect to the online depot ✓

Depot connection active
Proxy settings are not configured

[EDIT DEPOT CONNECTION](#)
[DISCONNECT](#)

Offline Depot ⓘ

Offline depot connection not set up. Start setup and enter the offline depot settings to setup connection.

[CONFIGURE](#)

Binary Management

Download Summary

Version: 9.0.0.0	VMware vSphere Foundation Downloaded	VMware Cloud Foundation Downloaded
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Filter the list by product and version and download the required component binaries. For additional guidance, refer to the [Product Interoperability Matrix](#)

Product VMware Cloud Foundation
Version 9.0.0

[DOWNLOAD](#)
[DELETE](#)

Deploy VMware Cloud Foundation

We will now begin the deployment of VMware Cloud Foundation.

1. Click '**DEPLOYMENT WIZARD**' to start the wizard.
2. Select '**VMware Cloud Foundation**' from the drop down menu as this is the cloud platform we will be deploying.

VMware Cloud Foundation Installer
Hello, admin@local

VMware Cloud Foundation Installer

VMware Cloud Foundation installer deploys and performs the initial configuration of a new VMware vSphere Foundation or VMware Cloud Foundation environment. The installation process automates the deployment and configuration of the various components in each solution.

[LEARN MORE](#)
[REVIEW PREREQUISITES](#)



Get Started

Download Binaries

Online depot connection active ✓

You can edit your depot connection and/or download additional software binaries

VMware vSphere Foundation 9.0.0.0 Downloaded
VMware Cloud Foundation 9.0.0.0 Downloaded

[DEPOT SETTINGS](#)

Deploy

You can deploy VMware Cloud Foundation or VMware vSphere Foundation using the deployment wizard or uploading a deployment specification JSON file.

[DEPLOYMENT WIZARD](#)
[DEPLOY USING JSON SPEC](#)

VMware Cloud Foundation
VMware vSphere Foundation

In this simulator, we will be deploying a new VCF Fleet. This option has been pre-selected. The VCF Installer can be used to deploy a VCF Instance in an existing VCF fleet as well.

1. Click on the vertical scroll bar on the bottom right to scroll down.
2. Click '**CONTINUE**' to begin.

The screenshot shows the 'Deploy VMware Cloud Foundation: What would you like to deploy?' screen. The interface has a dark blue header with 'VMware Cloud Foundation Installer' on the left and a user profile 'Hello, admin@local' on the right. Below the header, the title 'Deploy VMware Cloud Foundation: What would you like to deploy?' is displayed, followed by a link 'LEARN ABOUT VCF FLEET'. There are two main options, each with a diagram illustrating its components:

- Deploy a new VCF fleet** (selected with a radio button): The diagram shows a 'VCF Fleet' container. Inside, there are 'VCF Operations' and 'VCF Automation' components. Below these is a 'First VCF Instance' container, which contains a 'Management Domain' (with vCenter, NSX Manager, vSphere Cluster, and SDDC Manager) and 'Workload Domain(s)'. At the bottom of the fleet container is a dashed box for 'Additional VCF Instance(s)'.
- Deploy a VCF Instance in an existing VCF fleet** (unselected): The diagram shows an 'Existing VCF Fleet' container. It includes 'VCF Operations' and 'VCF Automation'. Below these is a text field for 'Existing VCF Instance(s)'. Then, it shows an 'Additional VCF Instance(s)' container, which contains a 'Management Domain' (with vCenter, NSX Manager, vSphere Cluster, and SDDC Manager) and 'Workload Domain(s)'. At the bottom is a dashed box for 'Additional VCF Instance(s)'.

At the bottom left, there is a note: 'If you intend to reuse existing components like a VMware vCenter or VCF Operations instance, subsequent steps will guide you to use them as building blocks in your deployment.' At the bottom right, there are two buttons: 'CANCEL' and 'CONTINUE'.

Existing Components

If you have existing VCF Operations or VMware vCenter already deployed, you can reuse them for this VCF Fleet deployment. In this simulator, we will be deploying new.

1. Click '**NEXT**' to proceed to the next step.

VMware Cloud Foundation Installer | Hello, admin@local

Deploy VCF Fleet

- Existing Components
- General Information
- VCF Operations
- VCF Automation
- vCenter
- NSX Manager
- Storage
- Hosts
- Networks
- Distributed Switch
- SDDC Manager
- Review
- Validate & Deploy

Existing Components

Select the existing components you want to reuse in this VCF fleet. You will be asked to provide details about the existing components later in the deployment wizard.

- VCF Installer validates the selected components to ensure that they are compatible.
- If you select a VCF Operations instance that is connected to a fleet management appliance or a VCF Automation instance, these components will also be added to this VCF fleet.
- It is recommended you select the VMware vCenter where the selected VCF Operations is currently deployed to be used as the vCenter for the first VCF instance.
- If a VMware NSX Manager is connected to the vCenter instance you intend to use, select the option to use existing NSX Manager.

☐ **VCF Operations**
 Use an existing VCF Operations instance as the management component in the VCF fleet.

☐ **VMware vCenter**
 Use an existing vCenter as the management domain vCenter.

VCF Fleet
 VCF Operations
 VCF Automation
VCF Instance
 Management Domain
 vCenter
 NSX Manager
 vSphere Cluster
 SDDC Manager

CLOSE NEXT

General Information

We will begin filling out general information about this VCF deployment. The default and only version of VCF 9.0.0 has been preselected.

- Click on the **VCF Instance Name** text field.
- Type the name '**vcf-a**'.
- Click on the **Management domain name** text field.
- Type the name '**mgmt-a**'.
- You have the option to choose between a single-node or three-node deployment model for the various VCF appliances. For this simulation, select '**Simple (Single-node)**'.
- Click on the **DNS domain name** text field.
- Type '**site-a.vcf.lab**'.
- Click on the **DNS servers** text field.
- Type '**10.1.1.1**' for this lab.

VMware Cloud Foundation Installer Hello, admin@local

Deploy VCF Fleet

- 1 Existing Components
- 2 General Information
- 3 VCF Operations
- 4 VCF Automation
- 5 vCenter
- 6 NSX Manager
- 7 Storage
- 8 Hosts
- 9 Networks
- 10 Distributed Switch
- 11 SDDC Manager
- 12 Review
- 13 Validate & Deploy

General Information

Enter the general configuration details for this deployment. All fields marked with a * are required.

Version * 9.0.0.0
The latest downloaded version of each component will be used

VCF Instance Name * vcf-a
Descriptive name for the VCF instance

Management domain name * mgmt-a
This name will also be used to generate names for objects in the management domain, like the vCenter, distributed switches etc.

Advanced: Custom networking configuration for VCF Operations and VCF Automation

☐ I want to connect VCF Operations and VCF Automation instances on different DVPs or NSX segments
By default, the VCF Installer configures all VCF Operations and VCF Automation appliances to use the same distributed virtual port group (DVP). To specify a different configuration, select the checkbox and perform a custom deployment of VCF Operations and VCF Automation after VCF Installer deploys the other components. [Learn more](#)

Deployment model *

☒ Simple (Single-node)
☐ High Availability (Three-node)
This selection only applies to newly deployed appliances of VCF Operations, VCF Automation and NSX Manager. Existing appliances that you plan to use in your VCF will not be modified. [Learn more](#)

DNS and NTP servers
This selection only applies to newly deployed appliances.

DNS domain name * site-a.vcf.lab

DNS servers * 10.1.1.1
Enter multiple DNS servers as comma-separated.

VCF Fleet

VCF Operations

VCF Automation

VCF Instance

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

[CLOSE](#) [BACK](#) [NEXT](#)

10. Click on the vertical scroll bar to the right to scroll down to view more.
11. Click on the '**NTP servers**' text field.
12. Enter '**10.1.1.1**' for this lab.
13. You have the option to specify your own passwords for the newly installed appliances or let the system generate.
For this simulation, check the box '**Auto-generate passwords for newly installed appliances**'.
14. Click '**NEXT**' to proceed to the next section.

VMware Cloud Foundation Installer | Hello, admin@local

Deploy VCF Fleet

- 1 Existing Components
- 2 General Information**
- 3 VCF Operations
- 4 VCF Automation
- 5 vCenter
- 6 NSX Manager
- 7 Storage
- 8 Hosts
- 9 Networks
- 10 Distributed Switch
- 11 SDDC Manager
- 12 Review
- 13 Validate & Deploy

General Information

ADVANCED: Custom networking configuration for VCF Operations and VCF Automation

☐ I want to connect VCF Operations and VCF Automation instances on different DVPGs or NSX segments

By default, the VCF Installer configures all VCF Operations and VCF Automation appliances to use the same distributed virtual port group (DVPG). To specify a different configuration, select the checkbox and perform a custom deployment of VCF Operations and VCF Automation after VCF Installer deploys the other components. [Learn more](#)

Deployment model *

This selection only applies to newly deployed appliances of VCF Operations, VCF Automation and NSX Manager. Existing appliances that you plan to use in your VCF will not be modified. [Learn more](#)

☒ Simple (Single-node)

☐ High Availability (Three-node)

DNS and NTP servers

This selection only applies to newly deployed appliances.

DNS domain name * [?](#)

DNS servers *
Enter multiple DNS servers as comma-separated.

NTP servers *
Enter multiple NTP servers as comma-separated.

Password creation

☒ Auto-generate passwords for newly installed appliances

Other

☒ Enable Customer Experience Improvement Program (CEIP)

CLOSE BACK **NEXT**

VCF Operations

This section asks for information related to the VCF Operations appliance deployment. As we had specified a 'Simple (Single-Node)' deployment model previously, only information for a single-node deployment is requested.

1. Click on the drop-down list to choose the **Operations Appliance Size**.
2. You have a choice between Extra Small, Small, Medium, Large and Extra Large. Click on '**Small**'.
3. Click on the **Operations primary FQDN** text field.
4. Type '**ops-a.site-a.vcf.lab**'.
5. Click on the Fleet Management Appliance **Appliance FQDN** text field.
The Fleet Management Appliance is an appliance that is deployed to manage deployments and lifecycle management of VCF Fleet components.
6. Type '**opslcm-a.site-a.vcf.lab**'.
7. Click on the Operations Collector Appliance **Appliance FQDN** text field.
8. Enter '**opscollector-01a.site-a.vcf.lab**'.
9. Click '**NEXT**' to proceed to the next section.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

1 Existing Components

2 General Information

3 VCF Operations

4 VCF Automation

5 vCenter

6 NSX Manager

7 Storage

8 Hosts

9 Networks

10 Distributed Switch

11 SDDC Manager

12 Review

13 Validate & Deploy

VCF Operations

Enter the configuration details for the VCF Operations appliances. All fields marked with a * are required.

VCF Operations Appliance

Operations Appliance Size *
Small

4 vCPUs and 16GB Memory

Operations primary FQDN *
ops-a.site-a.vcf.lab

Fleet Management Appliance

Appliance FQDN *
opsfcm-a.site-a.vcf.lab

Operations Collector Appliance

Provide details for deploying a new operations collector appliance for this VCF instance.

Appliance FQDN *
opscollector-01a.site-a.vcf.lab

VCF Fleet

VCF Operations

VCF Automation

VCF Instance

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

CLOSE

BACK

NEXT

VCF Automation

This section asks for information related to the VCF Automation appliance deployment. As we had specified a 'Simple (Single-Node)' deployment model previously, only information for a single-node deployment is requested. If you have an existing Aria Automation instance and wish to use it instead of deploying new, you can select the checkbox 'I want to connect a VCF Automation instance later'. For this simulation, we will deploy a new appliance.

1. Click on the **Appliance FQDN** text field.
2. Type '**auto-a.site-a.vcf.lab**'.
3. Click on the **Node IP 1** text field.
4. Type '**10.1.1.71**'.
5. Click on the **Node IP 2** text field.
6. Type '**10.1.1.72**'.
7. Click on the **Node name prefix** text field.
8. Type '**auto-a**'.
9. Click '**NEXT**' to proceed to the next section.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

1 Existing Components

2 General Information

3 VCF Operations

4 VCF Automation

5 vCenter

6 NSX Manager

7 Storage

8 Hosts

9 Networks

10 Distributed Switch

11 SDDC Manager

12 Review

13 Validate & Deploy

vCenter

Enter the configuration details for the management domain vCenter. All fields marked with a * are required.

Appliance FQDN *

vc-mgmt-a.site-a.vcf.lab

Appliance Size *

Small

Appliance Storage Size

Default

Datacenter Name *

mgmt-a-dc01

Cluster Name *

mgmt-a-cl01

SSO Domain Name

vsphere.local

VCF Fleet

VCF Operations

VCF Automation

VCF Instance

Management Domain

vCenter

NSX Manager

vSphere Cluster

SDDC Manager

CLOSE

BACK

NEXT

NSX Manager

This section asks for information related to the NSX Manager appliance deployment. As we had specified a 'Simple (Single-Node)' deployment model previously, only information for a single-node deployment is requested. You have a choice of NSX Manager appliance sizes to choose from. We will choose the pre-selected default size of Medium.

1. Click on the **Cluster FQDN** text field.
2. Type '**nsx-mgmt-a.site-a.vcf.lab**'.
3. Click on the **Appliance FQDN** text field.
4. Type '**nsx-mgmt-01a.site-a.vcf.lab**'.
5. Click '**NEXT**' to proceed to the next section.

VMware Cloud Foundation Installer Hello, admin@local

Deploy VCF Fleet

1 Existing Components

2 General Information

3 VCF Operations

4 VCF Automation

5 vCenter

6 NSX Manager

7 Storage

8 Hosts

9 Networks

10 Distributed Switch

11 SDDC Manager

12 Review

13 Validate & Deploy

NSX Manager

Enter configuration details for the NSX Manager to be deployed. All fields marked with a * are required.

Appliance Size *

Medium

6 vCPU, 24GB RAM, 300GB Storage

Cluster FQDN *

nsx-mgmt-a.site-a.vcf.lab

Appliance FQDN *

nsx-mgmt-01a.site-a.vcf.lab

VCF Fleet

VCF Operations

VCF Automation

VCF Instance

Management Domain

VCenter

NSX Manager

vSphere Cluster

SDDC Manager

CLOSE

BACK

NEXT

Storage

This section asks for information related to principal storage that will be used for the management domain. You have a choice of vSAN, VMFS on Fibre Channel (FC) or NFS v3.

1. Click on the **vSAN Architecture** drop down list.
2. You have a choice of vSAN ESA or vSAN OSA. For this simulation, click on '**vSAN OSA**'.

We will leave all the other options with auto-generated defaults. Take a moment to observe the customizable options available.

1. Click '**NEXT**' to proceed to the next section.

Hosts

This section asks for information related to ESX hosts that will be used for the management domain. For this simulation, we will be deploying with four hosts.

1. Click on the **ESX Root Password** text field to enter the password.
2. Click '**ADD HOST**' to add a fourth ESX host entry.
3. Click on the '**enter FQDN**' text field of the first host.
4. Type '**esx-01a.site-a.vcf.lab**'.
5. Click '**ADD**' to add the host.
6. Click on the '**enter FQDN**' text field of the second host.
7. Type '**esx-02a.site-a.vcf.lab**'.
8. Click '**ADD**' to add the host.
9. Click on the '**enter FQDN**' text field of the third host.
10. Type '**esx-03a.site-a.vcf.lab**'.
11. Click '**ADD**' to add the host.
12. Click on the '**enter FQDN**' text field of the fourth host.
13. Type '**esx-04a.site-a.vcf.lab**'.
14. Click '**ADD**' to add the host.
15. Click '**CONFIRM ALL FINGERPRINTS**' to confirm the SSL certificate fingerprint of all hosts.

16. Click '**NEXT**' to proceed to the next section.

VMware Cloud Foundation Installer | Hello, admin@local

Deploy VCF Fleet

- 1 Existing Components
- 2 General Information
- 3 VCF Operations
- 4 VCF Automation
- 5 vCenter
- 6 NSX Manager
- 7 Storage
- 8 Hosts**
- 9 Networks
- 10 Distributed Switch
- 11 SDDC Manager
- 12 Review
- 13 Validate & Deploy

Hosts

Select the hosts to add to the management domain of your VCF Instance. All fields marked with a * are required.

ESX Root Password *

Add Hosts

The minimum number of required hosts is displayed in the table. Click Add Host to add more ESX hosts.

	FQDN	Confirm Fingerprint
☐	esx-01a.site-a.vcf.lab	36 B6 F9 9E E6 6A AE 88 17 E6 90 34 C9 37 97 DA D4 2B 32 BA CB F3 06 F7 31 F9 6B 25 C6 C8 94 9D
☐	esx-02a.site-a.vcf.lab	87 DD 78 5F 0A BA 6B 8F 44 D8 E3 D1 79 81 88 11 E 9 8B 3A FD FD CD DB F2 69 8C 16 75 2E 10 AD DD
☐	esx-03a.site-a.vcf.lab	A8 C6 08 3F 64 F2 0B 01 CB 83 D3 A2 B7 16 87 7A AA 4C 4C A9 26 8F 7F BD FF 80 0D 0D BE 71 20 42
☐	esx-04a.site-a.vcf.lab	D1 DD 7B E2 9D 6F AC B7 FA DA 58 AC 88 25 AC 0 E 37 59 07 14 CD 80 C1 C5 A3 37 47 AF 14 AA AA 50

VCF Fleet

- VCF Operations
- VCF Automation

VCF Instance

Management Domain

- vCenter
- NSX Manager
- vSphere Cluster
- SDDC Manager

Networks

This section asks for information related to networks that will be used by the ESX hosts in the management domain. We will need the network information for ESX Management, VM Management, vMotion and vSAN. For this simulation, we will be using the same network for both ESX and VM Management.

ESX Management Network

1. Click on the **VLAN ID** text field.
2. Type '**10**'.
3. Click on the **CIDR Notation** text field.
4. Type '**10.1.1.0/24**'.
5. Click on the **Gateway** text field.
6. Type '**10.1.1.1**'.

VM Management Network

1. Select the check box **Use same inputs from ESX Management Network**.

vMotion Network

1. Click on the **VLAN ID** text field.
2. Type '**13**'.
3. Click on the **CIDR Notation** text field.
4. Type '**10.1.3.0/25**'.

5. Click on the **MTU** text field.
6. Delete and replace the text with **8000**.
7. Click on the **Gateway** text field.
8. Type '**10.1.3.1**'.
9. Click on the **IP Address Range** text field.
10. Type '**10.1.3.101**'.
11. Click on the '**To**' text field.
12. Type '**10.1.3.120**'.

vSAN Network

1. Click on the **VLAN ID** text field.
2. Type '**11**'.
3. Click on the **MTU** text field.
4. Delete and replace the text with **8000**.
5. Click on the **CIDR Notation** text field.
6. Type '**10.1.2.0/25**'.
7. Click on the **Gateway** text field.
8. Type '**10.1.2.1**'.
9. Click on the vertical scroll bar to the right to scroll down the page.
10. Click on the **IP Address Range** text field.
11. Type '**10.1.2.101**'.
12. Click on the '**To**' text field.
13. Type '**10.1.2.120**'.
14. Click '**NEXT**' to proceed to the next section.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

- 1 Existing Components
- 2 General Information
- 3 VCF Operations
- 4 VCF Automation
- 5 vCenter
- 6 NSX Manager
- 7 Storage
- 8 Hosts
- 9 Networks**
- 10 Distributed Switch
- 11 SDDC Manager
- 12 Review
- 13 Validate & Deploy

Networks

physical switches in your environment. All fields marked with a * are required.

ESX Management Network

VLAN ID * 10 MTU * 1500

CIDR Notation * 10.1.1.0/24 Gateway * 10.1.1.1

VM Management Network

☒ Use same inputs from ESX Management Network

vMotion Network

VLAN ID * 13 MTU * 8000

CIDR Notation * 10.1.3.0/25 Gateway * 10.1.3.1

IP Address Range * 10.1.3.101 To * 10.1.3.120

vSAN Network

VLAN ID * 11 MTU * 8000

CIDR Notation * 10.1.2.0/25 Gateway * 10.1.2.1

IP Address Range * 10.1.2.101 To * 10.1.2.120

VCF Instance

Management Domain

- ☒ vCenter
- ☒ NSX Manager
- ☒ vSphere Cluster
- ☐ SDDC Manager

CLOSE BACK **NEXT**

Distributed Switch

This section asks for information related to the distributed switch configuration that will be used by the management domain cluster. For this simulation, we have 4 pNics for each ESX host and we will configure them for NSX Traffic Separation where the NSX traffic will have its own dedicated pair of uplinks.

1. Click on the vertical scroll bar to the right to scroll down the page.
2. Click **'SELECT'** on the **NSX Traffic Separation** box to create 2 distributed switches. One for the NSX traffic and another for others.
3. We are now presented with configuration options for the distributed switches. Click on the **MTU** text field of **mgmt-a-cl01-vds02**.
4. Delete and replace the text with **8000**.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

- 1 Existing Components
- 2 General Information
- 3 VCF Operations
- 4 VCF Automation
- 5 vCenter
- 6 NSX Manager
- 7 Storage
- 8 Hosts
- 9 Networks
- 10 Distributed Switch**
- 11 SDDC Manager
- 12 Review
- 13 Validate & Deploy

Distributed Switch

Provide the vSphere Distributed Switch configuration to be applied to the hosts in the cluster.

4 physical network adapters are available to use. [View Details](#)

Selected Profile: NSX Traffic Separation [CHANGE](#)

This profile creates two distributed virtual switches with separate physical NICs. One switch is dedicated for NSX Edge and overlay traffic while the other is used for all other traffic.

Distributed Switches [RESET CHANGES](#)

Distributed Switch Name
mgmt-a-cl01-vds01
mgmt-a-cl01-vds02

mgmt-a-cl01-vds02

Distributed Switch Name: mgmt-a-cl01-vds02

MTU: 8000

Number of Uplinks: 2

Map uplinks to physical network adapters on host:

uplink1	vmnic2
uplink2	vmnic3

Network Traffic: NSX

☒ Apply default operation mode configured in NSX Manager

VCF Fleet

- VCF Operations
- VCF Automation

VCF Instance

Management Domain

- vCenter
- NSX Manager
- vSphere Cluster
- SDDC Manager

CLOSE BACK **NEXT**

- Click on the vertical scroll bar to the right to scroll down the page.
- Click on the **NSX-Overlay VLAN ID** text field.
- Type '14'.
- Select '**DHCP**' for the **IP Assignment (TEP)** option.
- Click '**NEXT**' to proceed to the next section.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

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- 4 VCF Automation
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- 6 NSX Manager
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- 9 Networks
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Distributed Switch

Number of Uplinks: 2

Map uplinks to physical network adapters on host:

uplink1	vmnic2
uplink2	vmnic3

Network Traffic: NSX

☒ Apply default operation mode configured in NSX Manager

Load Balancing: Route based on the source of the port ID

uplink1: ☒ Active ☐ Standby ☐ Unused

uplink2: ☒ Active ☐ Standby ☐ Unused

Transport Zones

NSX-Overlay

Transport Zone Name	VCF-Created-Overlay-Zone
VLAN ID	14
IP Assignment (TEP)	☒ DHCP ☐ IP Pool

CLOSE BACK **NEXT**

SDDC Manager

This section asks for information related to the SDDC Manager appliance deployment.

1. Click on the **Appliance FQDN** text field.
2. Type '**sddcmanager-a.site-a.vcf.lab**'.
3. Click '**NEXT**' to proceed to the next section.

The screenshot shows the 'SDDC Manager' configuration step in the VMware Cloud Foundation Installer. On the left, a 'Deploy VCF Fleet' sidebar lists steps 1 through 13, with '11 SDDC Manager' highlighted. The main area is titled 'SDDC Manager' and contains a message: 'Enter the configuration details for the SDDC Manager appliance. All fields marked with a * are required.' Below this, a note states: 'The VMware Cloud Foundation installer appliance is not deployed on one of the hosts in the management domain. During the deployment process, a new SDDC Manager appliance will be deployed.' The 'Appliance FQDN *' field is populated with 'sddcmanager-a.site-a.vcf.lab'. A tooltip explains: 'Either a Fully Qualified Domain name e.g host.vsphere.local or just the hostname e.g. host'. On the right, a 'VCF Instance' summary box shows a 'Management Domain' containing 'vCenter', 'NSX Manager', 'vSphere Cluster', and 'SDDC Manager'. At the bottom right are 'CLOSE', 'BACK', and 'NEXT' buttons.

Review

We are now ready to review the information entered before we proceed to validate and deploy VCF. Take a moment to review the information entered.

1. Click on the vertical scroll bar to the right to scroll down the page to see more.
2. When ready, click '**NEXT**' to proceed to the next section.

VMware Cloud Foundation Installer

Hello, admin@local

Deploy VCF Fleet

1 Existing Components

2 General Information

3 VCF Operations

4 VCF Automation

5 vCenter

6 NSX Manager

7 Storage

8 Hosts

9 Networks

10 Distributed Switch

11 SDDC Manager

12 Review

13 Validate & Deploy

Review

▼ Distributed Switch	mgmt-a-cl01-vds02
MTU	8000
▼ Uplinks mappings	
vmnic2	uplink1
vmnic3	uplink2
▼ Network traffic	
▼ NSX	
Operational Mode	Apply default operation mode configured in NSX Manager
Load Balancing	Route based on the source of the port ID
Active Uplinks	uplink1, uplink2
▼ Transport Zone Name	VCF-Created-Overlay-Zone
VLAN ID	14
IP Assignment (TEP)	DHCP
▼ SDDC Manager	
Appliance FQDN	sddcmanager-a.site-a.vcf.lab
VCF Installer Deployment	New SDDC Manager Deployment

CLOSE

BACK

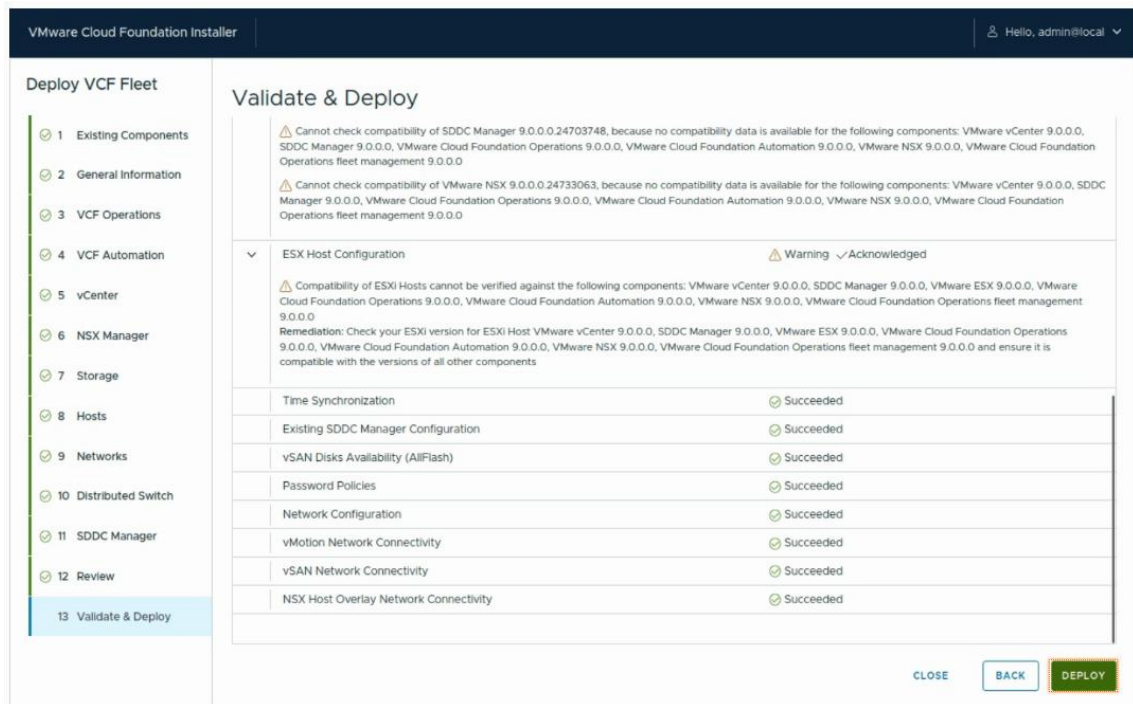
NEXT

Validate & Deploy

VCF Installer will automatically validate the information entered.

Note: As this is a simulation lab, there will be a few benign warnings that can be safely ignored.

1. Click **ACKNOWLEDGE ALL WARNINGS** to ignore the warnings. All other validations should succeed with no errors.
2. Click on the vertical scroll bar to the right to scroll down the page to see more.
3. Click **DEPLOY** to proceed deploying VMware Cloud Foundation.



SUCCESS!

You have successfully deployed VMware Cloud Foundation. Let's review what had happened.

1. Click on the **Expand** arrows to view some of the tasks completed. The VCF Installer has successfully executed hundreds of tasks to deploy VCF automatically based on the inputs you had previously entered.
2. Click on the **Shrink** arrows to minimize the tasks view.
3. Click on the vertical scroll bar to the right to scroll up the page.
4. As we selected to auto-generate passwords, we can view the passwords generated for each of the components installed. Click **REVIEW PASSWORDS**.

VMware Cloud Foundation Installer

Hello, admin@local

RETURN HOME



Congratulations! Your deployment completed successfully!

You can now follow the next steps recommendations described in the right panel

DOWNLOAD JSON SPEC

REVIEW PASSWORDS

100% completed

Deploy vCenter

Deploy SDDC Manager

Configure the vSphere cluster

Deploy and configure NSX



Next Steps

1. Log in to the VCF Operations UI.

2. Navigate to License Management, and apply your VCF license keys within the 60-day evaluation period.

VCF Operations Login

[ops-a-site-a-vcf-lab](#)

Username: admin

Password: *****

Tasks

VMware Cloud Foundation Deployment

Subtask

Description

Status

Start time

Last updated

Record VCF Installer Workflow Execution completion in Customer Experience Improvement Program (CEIP)

Record VCF Installer Workflow Execution completion in Customer Experience Improvement Program (CEIP)

Success

5/27/25, 3:25 AM

5/27/25, 3:25 AM

Updating deployment status of VCF Automation to succeeded

Updating deployment status of VCF Automation to succeeded

Success

5/27/25, 3:25 AM

5/27/25, 3:25 AM

The FQDN and credentials for each of the appliances installed can be viewed from this screen. The passwords can be changed from VCF Operations as a day 2 operation.

This concludes the VCF Installation simulation.